

Renewable Energy Statistics 2025

Global Economic Cooperation and Development (GECD) · 2025-04

1. Scope and definitions

This edition of Renewable Energy Statistics covers the twenty-nine member economies of the Global Economic Cooperation and Development (GECD), together with four accession candidates whose data are reported separately in the annex tables and are excluded from all aggregates presented in the main text. The reference period is the calendar year 2024, with provisional quarterly estimates extending to the fourth quarter of 2024 for members participating in the accelerated reporting facility. Data were compiled by the GECD Energy Statistics Division from national submissions received through the harmonised Annual Energy Questionnaire between June and December 2024, supplemented by administrative registries of grid-connected installations. Where national submissions were incomplete, the Secretariat applied the estimation procedures described in the methodological annex; imputed values account for 3.2% of reported generation and 1.9% of reported capacity across the member group.

The statistical unit throughout is the generating installation, defined as a physically contiguous set of generating equipment operating under a single grid connection agreement. Renewable sources comprise hydropower, wind, solar photovoltaic, concentrated solar thermal, geothermal, tidal and wave, and the biogenic fraction of solid, liquid and gaseous bioenergy. Pumped storage hydro is recorded separately and is not counted as renewable generation, since its output derives from the mix of the charging electricity rather than from a primary renewable flow. Municipal waste is apportioned between renewable and non-renewable components using the fifty-fifty default convention unless a member reports a measured biogenic fraction, which twelve members now do.

Two conventions deserve particular emphasis because they are the most frequent source of misreading in secondary use of this publication. Shares in this publication are expressed as a percentage of gross electricity generation, not of primary energy supply. Gross generation is measured at the generator terminals before deduction of station service consumption, which averaged 2.4% of gross output across thermal renewable plants in 2024; net generation figures, where required, are given only in Table A.3 of the annex. Users comparing these shares with figures published on a primary energy basis should expect systematic divergence, since the substitution method applied to non-combustible renewables in primary energy accounting produces materially different denominators.

Capacity is reported as net maximum electrical capacity under standard reference conditions, expressed in megawatts. For solar photovoltaic, members report capacity in megawatts of alternating current at the inverter output; where national registries record only direct-current module ratings, the Secretariat applies a country-specific conversion ratio derived from the sample of installations for which both values are known, with a median ratio of 1.21 across the member group. Distributed installations below 1 MW are included where a national registry exists; six members are unable to report behind-the-meter capacity below 30 kW, and the resulting undercount is estimated at between 1.5% and 4.0% of national solar capacity in those cases. All monetary values in the accompanying investment tables are expressed in constant 2020 international dollars converted at purchasing power parities.

2. Capacity additions

Gross capacity additions across the member group totalled 412.7 GW in 2024, an increase of 18.3 % on the 348.9 GW added in 2023 and the fifth consecutive annual record in the series.

Retirements and repowering-related decommissioning removed 21.4 GW, giving net additions of 391.3 GW. The distribution of additions across technologies continued to narrow rather than broaden.

Solar photovoltaic accounted for 71% of all new capacity added across GECD members in 2024. Onshore wind contributed a further 17.8%, offshore wind 6.1%, and all remaining renewable technologies together the balance of just over five per cent.

The concentration of additions in a single technology reflects both cost dynamics and the shorter development lead times characteristic of ground-mounted photovoltaic projects, which averaged 26 months from permit application to commissioning in 2024 compared with 51 months for onshore wind and 89 months for offshore wind. Volume-weighted capital expenditure for utility-scale photovoltaic fell to 612 international dollars per kilowatt, 9.4% below the 2023 level, while onshore wind capital expenditure was essentially flat at 1,340 dollars per kilowatt. Members should note that the capital expenditure series was rebased in the 2024 wave to exclude grid connection charges levied by system operators, which were previously included by eleven of twenty-nine reporting members and excluded by the remainder; the rebased series is not directly comparable with figures published before the 2023 edition.

Bioenergy additions declined for the third consecutive year, falling to 4.9 GW from 7.1 GW in 2023 and 11.6 GW in 2021, as feedstock cost pressures and the expiry of legacy support schemes in several members reduced the pipeline of new combustion capacity. Geothermal additions of 1.2 GW were concentrated in three members, and hydropower additions of 8.3 GW were the lowest recorded since the series began in 2009, with two thirds of that total accounted for by capacity uprates at existing facilities rather than new impoundments. Tidal and wave capacity remains at demonstration scale, with cumulative installed capacity of 0.44 GW across the member group.

Users should interpret annual addition figures with care where commissioning dates straddle a year boundary. Members apply differing conventions for the date of first synchronisation versus the date of commercial operation, and the Secretariat estimates that between 6 GW and 14 GW of capacity recorded in 2024 could plausibly have been assigned to 2023 or 2025 under alternative conventions. A harmonised commissioning definition has been agreed by the Working Party on Energy Statistics and will take effect from the 2027 wave, at which point the historical series will be revised back to 2018. Comparisons at the level of individual members and individual technologies are more sensitive to this issue than aggregate comparisons, since offsetting errors partially cancel in the group total.

3. Cross-country comparison

Cross-country comparison of renewable penetration is presented here on a consistent gross generation basis for all members, which in several cases produces figures that differ from those published in national statistical yearbooks. The GECD average share of renewables in electricity generation reached 34.6% in 2024. This is a generation-weighted average across the member group rather than an unweighted mean of national shares; the unweighted mean, reported for completeness, was 41.2%, reflecting the fact that several smaller members with abundant hydropower resources record shares well above the group aggregate while the largest generating

economies remain below it. The gap between the two measures has widened by 1.4 percentage points since 2020 and should be borne in mind whenever a single headline figure is quoted.

Dispersion across members remains substantial. The interquartile range of national shares ran from 23.7% to 58.1%, and the ratio between the highest and lowest national shares was 6.8 to one, down from 8.2 to one in 2019. Eleven members recorded shares above fifty per cent, of which eight rely predominantly on hydropower endowments rather than on variable renewable deployment. Four members recorded year-on-year declines in their renewable share despite positive capacity additions, in each case attributable to below-average hydrological conditions rather than to changes in installed capacity; hydropower output across the group fell 3.9% on 2023 despite a larger installed base.

The relationship between capacity share and generation share continues to loosen as the technology mix shifts. Variable renewable technologies accounted for 61.4% of installed renewable capacity across the group but only 44.8% of renewable generation, a divergence driven by capacity factors that averaged 14.9% for solar photovoltaic, 29.6% for onshore wind and 43.1% for offshore wind, against 51.7% for hydropower and 68.2% for geothermal. Analysts constructing rankings from capacity data alone will therefore reach materially different conclusions from those working with generation data. The Secretariat recommends that generation-based measures be used for any comparison intended to characterise the composition of electricity supply.

Curtailment reporting was expanded in this wave and now covers twenty-two members, up from fifteen in the 2023 edition. Reported curtailment of renewable output averaged 3.1% of available variable renewable energy, ranging from below 0.5% in members with strong interconnection to 11.4% in one member with substantial transmission constraints in its principal generating region. Because the reporting member set changed between waves, the 2024 curtailment average is not comparable with the 2.6% published for 2023; a like-for-like recalculation restricted to the fifteen members reporting in both years gives 2.9% for 2024. Policy readers should treat curtailment levels as an indicator of network adequacy rather than of deployment success, since the two move in opposite directions over the short run.

Finally, users are reminded that all figures for 2024 carry provisional status and will be revised in the 2026 edition. Historical revision experience suggests that the group generation share is typically revised by less than 0.3 percentage points, while national figures for members with large distributed generation fleets have been revised by up to 1.8 percentage points in past waves. Revisions to capacity addition figures have been larger in both directions, reflecting the commissioning date conventions discussed above. The complete revision history for each series is maintained in the online data warehouse accompanying this publication.